Shayan Shakeri

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PROFESSIONAL SUMMARY

AI Research Engineer working on vision-language-action (VLA) policies and visuomotor manipulation, from real-robot data collection through sim-to-real deployment. Placed 8th in Stanford's BEHAVIOR-1K manipulation challenge and open-sourced a flow-matching VLA integration for RLinf. Co-founder of Merlyn Labs; M.Eng in AI & Robotics, University of Toronto, expected April 2027.

CORE COMPETENCIES

VLA Policies | Behavior Cloning | Reinforcement Learning | Sim-to-Real | Grasp Planning | Agent Evaluation | PyTorch

WORK EXPERIENCE

Merlyn Labs
Aug 2025 - Present

AI Researcher & Co-Founder | Toronto, Canada

- Showed **π0.5 generalization failures on LIBERO-PRO are recipe-induced overfitting**; proposed an alternative baseline.
- Conservative finetuning doubled position-swap success (**21% to 42%**), showing spatial generalization over memorization.
- Developing **VLM judges** that score rollouts into dense, context-dependent RL rewards that are difficult to game.
- Open-sourced a **flow-matching VLA integration for RLinf**, enabling RL training on the BEHAVIOR-1K suite in OmniGibson.
- Running **sim-to-real transfer** of household tasks on a hand-built AlohaMini embodiment.

BMO AI Centre of Excellence
Sep 2025 - Present

AI Research Engineer | Toronto, Canada

- **Uncovered systematic bias to downplay investment risk** in a GenAI tool serving $200B+ AUM wealth management.
- Built a **deterministic agent eval pipeline** using hundreds of synthesized inputs to detect misaligned outputs at scale.
- Designing evals for LLM agents that retrieve and reason over commercial banking and insurance policy.
- Building a graph-based agentic system answering complex multi-hop relational queries across bank client data.
- Side project: repairing and reprogramming BMO's legacy branded greeter robot.

Epineuron Technologies
May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op | Mississauga, Canada

- Helped develop **PeriPulse**, an FDA Breakthrough-designated device now in multi-national clinical trials; designed and assembled PCBs and authored IEC 60601-1 / ISO 13485 validation protocols.
- Optimized power draw via oscilloscope and power-analyzer testing, achieving **900% battery life improvement**.

PROJECTS

Stanford BEHAVIOR-1K Challenge
8th Place, Standard Track

Trained VLA manipulation policies on 10,000+ demonstrations covering 22 of 50 scored task types. Identified **proprioceptive collapse** as a critical failure mode (60% masking improved task success up to 48%); found chunked execution outperformed temporal ensembling **3x**, exposing missing temporal awareness. Published a technical report and LessWrong analysis.

Voice-Controlled Robotic Prosthetic

Developed an LLM-based control pipeline converting natural language into manipulation sequences for a 7-DOF simulated arm. Integrated YOLO object detection with LiDAR depth mapping for 3D localization and grasp planning.

EDUCATION

M.Eng in AI & Robotics, University of Toronto — Sep 2024 - Apr 2027 (anticipated)

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics.

B.Eng, Engineering Physics Co-op, McMaster University — Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems.

SKILLS

Robot Learning & Manipulation: VLA Policies, Behavior Cloning, Reinforcement Learning, Imitation Learning, Grasp Planning, Sim-to-Real, Contact-Rich Manipulation

Simulation, Robotics & Languages: OmniGibson, MuJoCo, Flow Matching, ROS2, Perception (YOLO, LiDAR), PCB Design, Python, C/C++, PyTorch, JAX